Diwakar Sandhu

CS 4804 – Data Visualizations

7 March 2026

A Day in the Life of WPI Students

Overview and Motivation

"A Day in the Life of WPI Students" is an interactive data visualization project that explores how students at Worcester Polytechnic Institute spend their time across a typical day. College schedules can vary widely depending on classes, studying, work, and extracurricular activities, and it can be hard to understand the overall rhythm of student life beyond personal anecdotes. By collecting anonymous time-use schedules from WPI students and visualizing them across the 24-hour day, this project aims to reveal common patterns (like when students are most often in class, studying, eating, or sleeping) and show how routines differ across groups such as class year, living situation, and campus involvement.

Related Work

This project is inspired by FlowingData's "A Day in the Life of Americans," which visualizes daily routines as activities changing over time to show broad patterns in how people spend their day. That work demonstrates how time-use data can be compelling when presented as an evolving, temporal view rather than a static summary. This project builds on this idea by applying a similar time-based visualization approach to a WPI-specific dataset collected through a survey, with an emphasis on enabling exploration and comparison between different subsets of students.

Questions

This project investigates the following questions: What does a "typical day" look like for WPI students overall, and what activities dominate different times of day? When are students most likely to be sleeping, attending class, studying, working, or participating in leisure and extracurriculars? How do these patterns change across different groups, such as year, living situation, and involvements? Finally, what variability exists in routines and do most students follow a similar daily rhythm, or are there multiple distinct "day types" within the WPI community?

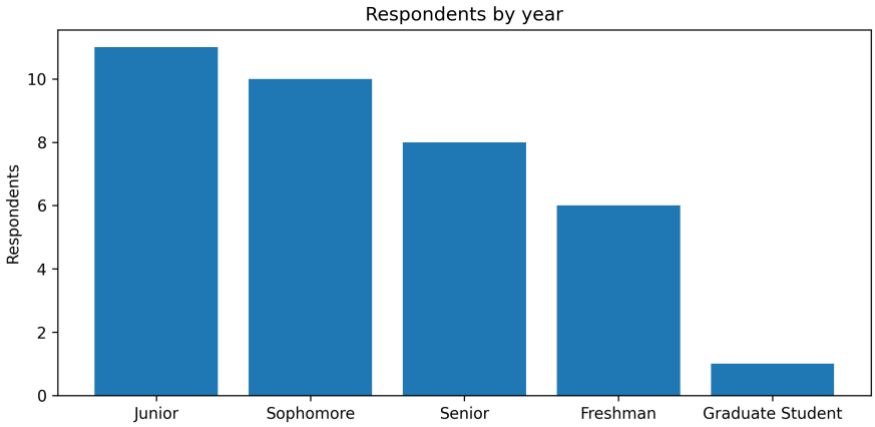
Data

To collect data for this project, I created a Google Form and distributed it to WPI students through friends, group chats, and other campus connections. The form began with a short set of demographic questions to support comparison across groups, including respondents' major(s), living situation, and campus involvements. The main portion of the form was a multi-grid section covering each hour of a full day, where participants selected a single activity that best described what they were primarily doing during that hour. Activity options included categories such as sleeping, personal care, eating, lecture/lab, studying, work, chores/errands, leisure, exercise/practice, etc.

In total, I received 36 responses. While this was enough to begin exploring patterns, the sample size was smaller than ideal for making broader claims about student life at WPI. In retrospect, I wish I had started collecting responses much earlier, since sharing the form over a longer period would likely have improved the return rate and produced a more representative dataset. I also made an effort to reach a wide range of students rather than only surveying people within my immediate social circles; however, the final set of responses still did not end up as balanced across groups as I had hoped.

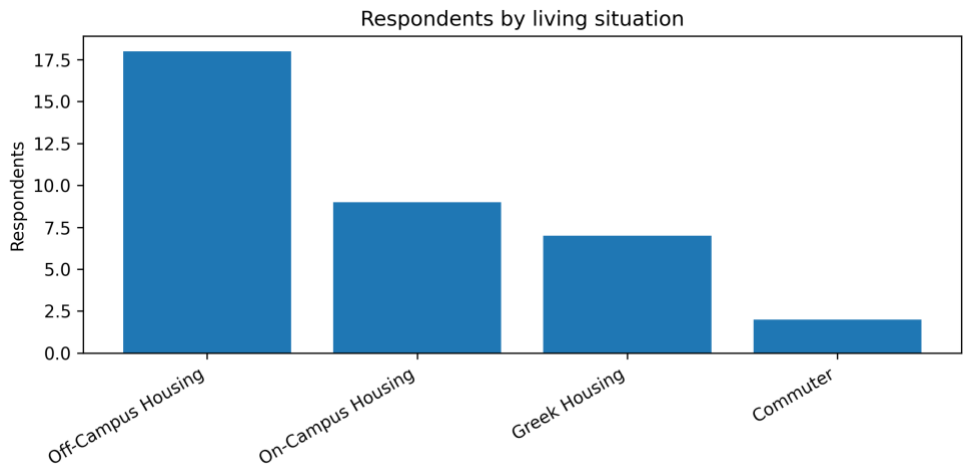
Exploratory Data Analysis

Before designing the final visualization, I did a quick exploratory analysis to understand who responded to the survey, check how balanced the sample was, and identify the any preliminary patterns. These initial plots helped me decide which comparisons would be meaningful and whether this dataset could be representative of the larger WPI community.

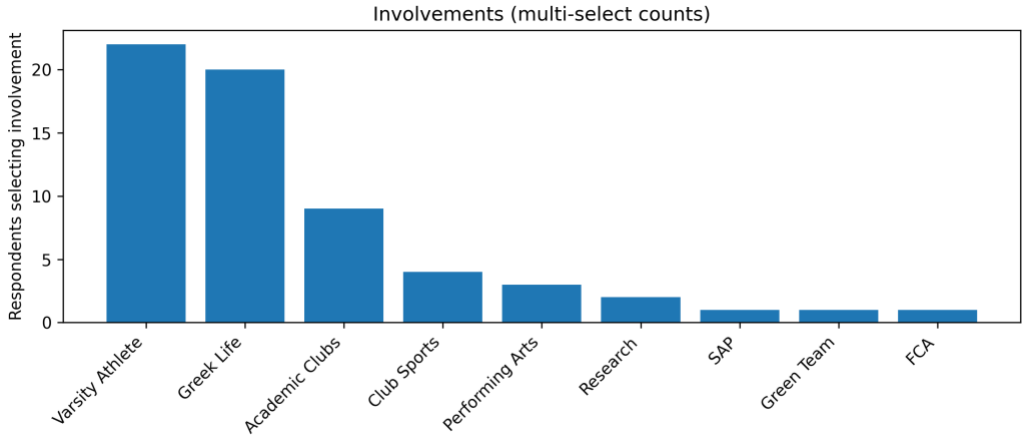


This chart shows that most responses came from upperclassmen, especially juniors and sophomores, with very few graduate students represented. This matters because class year can

strongly affect schedules (course load, project work, internships), so the overall “average day” in this dataset likely reflects upper-class routines more than the full WPI population.

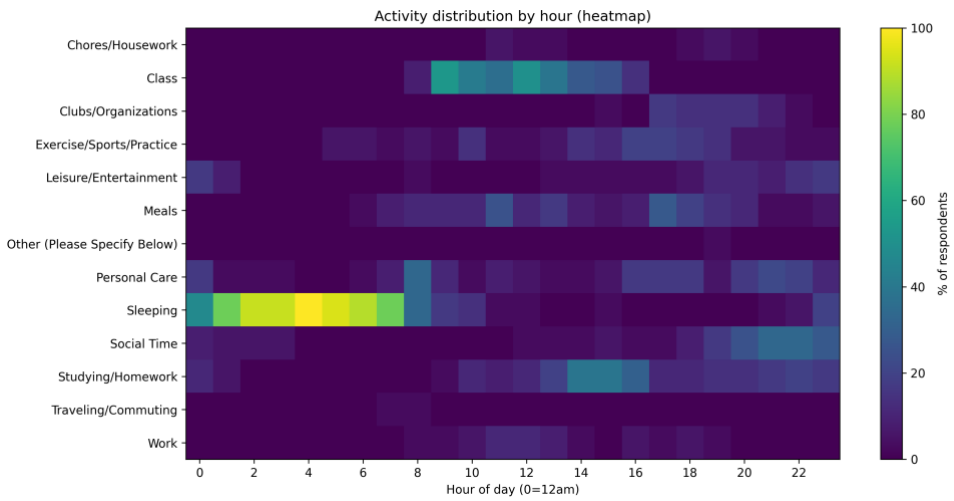


This sample skews heavily towards off-campus housing (which is common among upperclassmen), with fewer on-campus students (freshmen) and very few commuters. Because living situation affects commuting time, meal patterns, and campus involvement, this imbalance suggests, that any conclusions about commuting or on0campus dorm life should be treated with caution.

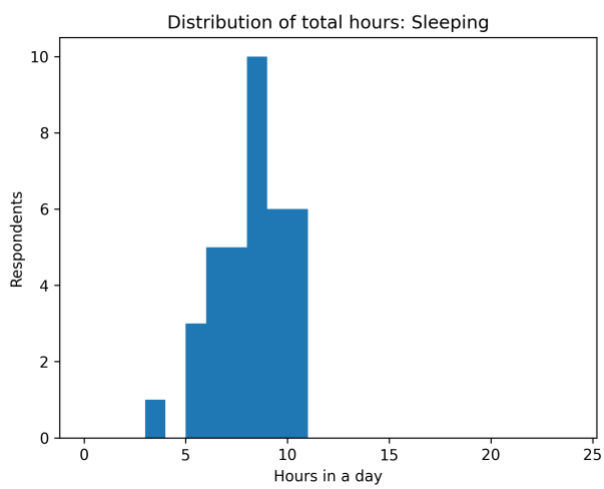


The involvement breakdown is dominated by varsity athletes and Greek like, while other categories (research, performing arts, some clubs) have smaller counts. This uneven distribution is important because it likely influences the activity patterns we see later, and it also limits how confidently we can compare smaller involvement groups. Therefore, any conclusions we make

must take into consideration that they might be more directed towards varsity athletes and Greek life.



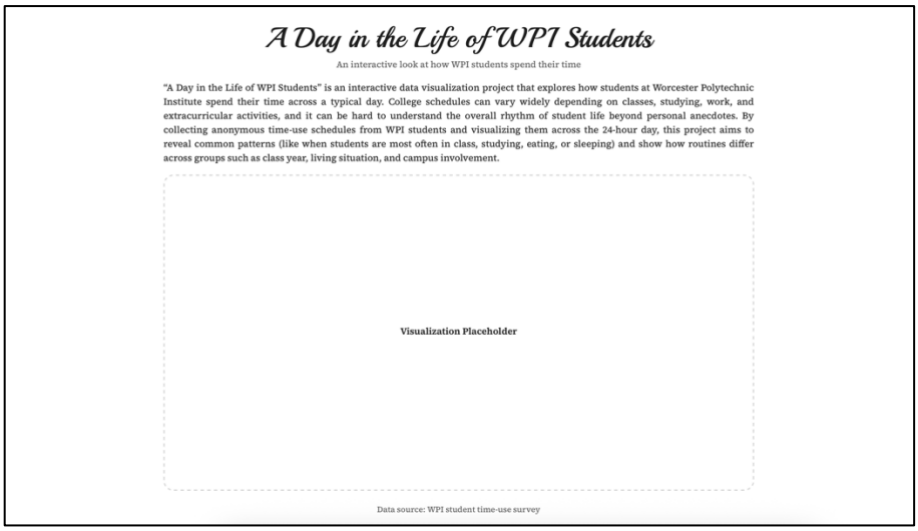
This heatmap captures the overall "shape of a typical day in the dataset: sleep dominates late night/early morning, class clusters around late morning to afternoon, and studying/social time appears more in the afternoon and evening. This plot confirmed that an hour-by-hour visualization will be effective since the strongest patterns are clearly time-dependent and visually distinct.



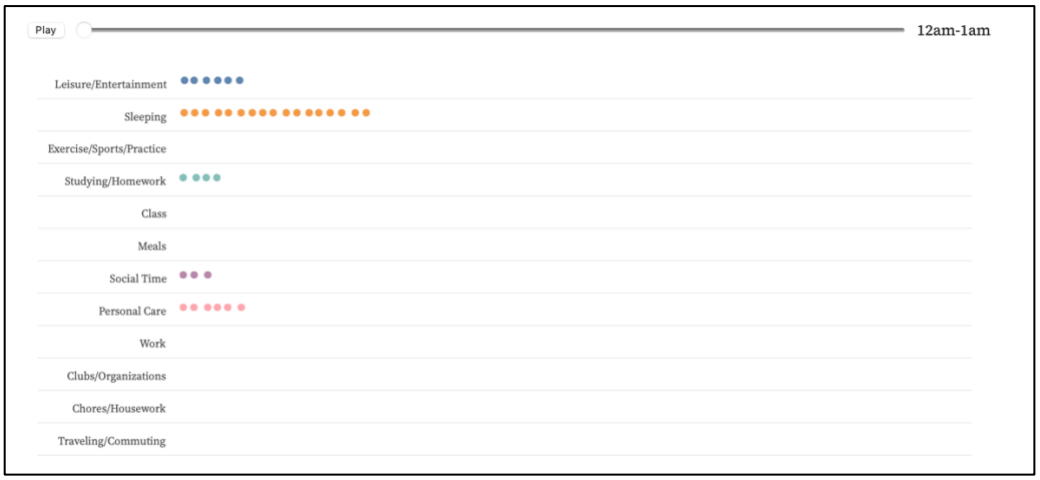
This histogram shows most respondents reporting roughly 7-10 hours of sleep, with a few sleeping less, indicating both a common pattern and meaningful variation between individuals. Sleep is the largest time block for most people, so it becomes a natural anchor activity to highlight when describing overall daily structure.

Design Evolution

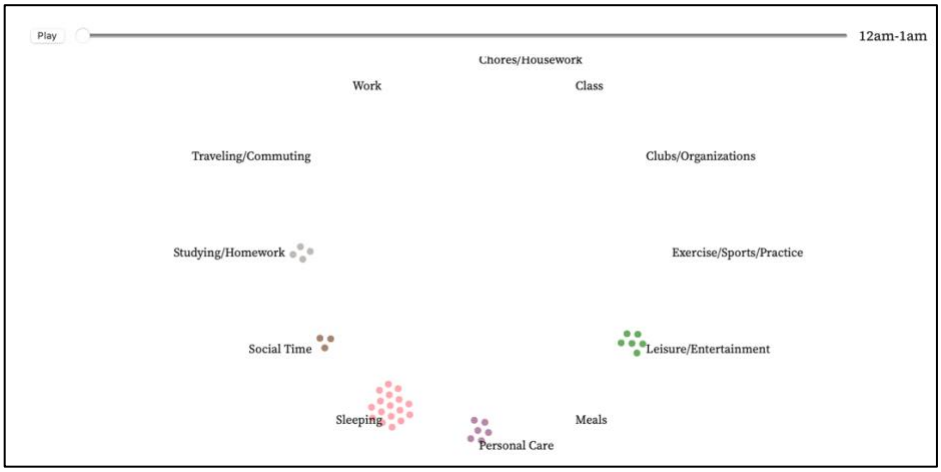
I started by building the overall structure of the project website before implementing any visualization logic. This version focused on having the centered script-style title, a short description explaining the motivation and goals, and a large framed placeholder where the visualization would eventually live. The goal here was to have a clean page hierarchy so later iterations could focus purely on interaction and animation.



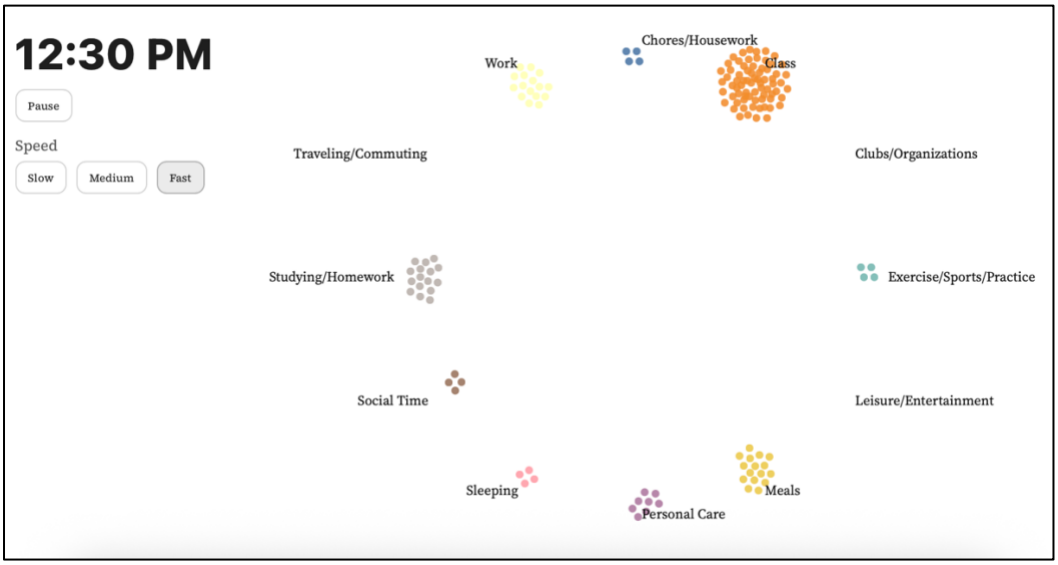
My first visualization attempt used a simple "lanes" design where each activity had its own horizontal row and each respondent was shown as a dot positioned within the row of their current activity. A play button and a time control allowed the dots to move hour-by-hour. This was effective for readability as viewers could quickly map dots to categories but it felt very static and didn't match the "flow" inspiration I was aiming for.



Next, I rearranged the activities into a circle and used a force-based animation so dots could “flow” between activity clusters as time advanced. This made transitions between activities much more apparent and created a more engaging sense of motion compared to the lane-based view. However, this iteration introduced new design challenges such as label overlap and dots not settling neatly into clusters, which required tuning forces and label placement.



In the final version, I redesigned the interface so the controls live on the left and the visualization fills the right side, improving clarity and making the interactive elements easier to find. I also replaced the hour slider with a large, bold time display and playback speed buttons (slow/medium/fast), which made the experience feel more like watching a day unfold rather than scrubbing through discrete steps. Even though the underlying data is hourly, presenting time continuously helped reinforce the storytelling goal of the project and made the animation feel smoother and more alive.



Implementation

The project was implemented as a single-page web application using HTML/CSS for layout and styling and D3.js for rendering and animation. The visualization loads a cleaned CSV dataset (clean_data.csv) containing one row per respondent and 24 columns representing the respondent's primary activity for each hour of the day.

The core visualization represents each respondent as a dot and arranges activity categories as fixed target positions around a circle. A D3 force simulation drives the animation: at each time step, dots are pulled toward the target corresponding to their current activity using forceX and forceY, while collision and mild repulsion forces dots from overlapped and helps clusters form cleanly. The interface includes play/pause controls and speed buttons (slow/medium/fast). The displayed time advances minute-by-minute, but because the survey data is hourly, a dot's activity only changes when the current minute crosses into a new hour; when that happens the simulation forces are updated and restarted to smoothly transition dots to their new activity cluster.

To support usability and presentation quality, the final site uses a two-panel layout with controls and a large time readout on the left and the visualization on the right. The visualization is also responsive so on window resizes, the circular activity targets and labels are recomputed to fit the updated dimensions while preserving the overall structure of the layout.

Evaluation

Overall, the visualization succeeds at providing an intuitive understanding of how students in the dataset move through a typical day. Watching dots flow between activity clusters makes daily rhythms easy to perceive such as how sleep dominates overnight, academic work becomes more common during the day, and schedules diversify in the evening with leisure, social time, and extracurriculars. Compared to static charts, the animated view especially effective for highlighting transitions between activities and showing how concentrated or spread-out students are at different times.

At the same time, the results should be interpreted as an exploratory snapshot, not a definitive representation of all WPI students. The dataset contains only 36 responses and the distribution is not fully balanced across groups (for example, it is skewed toward upperclassmen and students involved in Greek life and varsity athletics). Because of these limitations, the visualization is best used to generate hypotheses about student routines rather than to make strong claims about the entire campus population.

The project has strong potential to become a larger WPI-wide time-use experiment. With earlier and broader data collection, potentially though class-wide distribution, campus organizations, or institutional support, the visualization could capture a more representative sample and enable more meaningful comparisons across majors, years, and involvement times. A scaled-up version could help professors and administrators better understand how students balance class time, studying, sleep, work, and extracurriculars, and could inform decisions about workload, scheduling, and student support. Future improvements could also include collecting higher-resolution time blocks (30-minute intervals or start/end event blocks), incorporating weekday vs weekend differences, and adding filtering and comparison tools directly into the interface.